Karthickraja J

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.60
Nalandas Higher Secondary School	Class XII (Tamil Nadu State Board)	2015	98.50%
Nalandas Higher Secondary School	Class X (Tamil Nadu State Board)	2013	97.60%

PROJECTS

Nine-Switch Inverter for Reduced CMV with Increased Efficiency

- Proposed a hybrid Si + SiC nine switch VSI with reduced common mode voltage.
- Built and tested a 3kW prototype at 500V dc-link at 15kHz.
- Reduced peak-to-peak CMV from Vdc to Vdc/3, when compared with conventional VSI.
- Reduced the converter losses by 15% at 2.8kW.

Multi-phase Inverter for Drives with reduced CMV

- Designed and developed a hybrid inverter with Si and SiC switches for six phase systems.
- Built and tested a 3kW prototype at 300V dc-link at 5kHz on a six-phase induction motor.
- Reduced CMV to zero.

Position Control of BLDC Motor

- Implemented a position control of BLDC Motor using BOOSTXL-DRV8305EVM.
- A sensored control is implemented where the encoder data is communicated through SPI protocol.

Classification of Faults in a PV System

- Classified faults using a wavelet transform based approach
- Implemented the classification algorithm on a 5x3 PV array using NI-DAQ.

PATENTS

K. Jegatheesan and J. Titus, "A Voltage Source Inverter for reducing Common Mode Voltage," Indian Patent Application Number 202641099568. *(Under Examination)*

JOURNAL PUBLICATIONS

Arunachalam, V., **Karthickraja, J.** & Senthamizh Selvan, S. Wavelet transform based detection, classification and location of faults in a PV array. *J Ambient Intell Human Comput* **14**, 11227-11237 (2023). https://doi.org/10.1007/s12652-023-04628-3

CONFERENCE PUBLICATIONS

K. Jegatheesan and J. Titus, "Analytical Evaluation of Harmonic Performance of Space Vector based Common Mode Voltage Reduction PWM Techniques," *2026 Energy Conversion Congress & Expo Europe (ECCE Europe)*, Valencia, Spain, 2026, pp. 1-6. *(Accepted for Publication)*

K. Srinuprasad, K.K.R. Kumar, **K. Jegatheesan** and J. Titus, "A Modified Cascaded Dual Three-Phase Current Source Inverter for Common-Mode Voltage Reduction," *2026 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)*, Guwahati, India, 2026, pp. 1-6. *(Accepted for Publication)*

K. Jegatheesan and J. Titus, "A Comparative Evaluation of the Performance of a novel H9-VSI for Reduced Common Mode Currents," *2024 IEEE 33rd International Symposium on Industrial Electronics (ISIE)*, Ulsan, South Korea, 2024, pp. 1-6, doi: 10.1109/ISIE54533.2024.10595830.

K. Jegatheesan, S. S. S and V. Arunachalam, "Performance Analysis of a Solar-Fed Induction Motor Drive under Various PV Array Fault Conditions," *2022 International Conference for Advancement in Technology (ICONAT)*, Goa, India, 2022, pp. 1-6, doi: 10.1109/ICONAT53423.2022.9726058.

SKILLS

Simulation

- Inverter topologies, PWM algoithms, loss estimation.

PCB Design

- Built PCBs for Inverters, Gate Drivers and other Auxiliary Boards

Embedded/Control

- v/f control of Induction Motor, Position control of BLDC Motor

SCHOLASTIC ACHIEVEMENTS

Gold Medal in M.Tech Power Electronics, NIT Puducherry, 2022

RELEVANT COURSES

Power Converter Design

Modeling and Control of Electric Drives

Embedded System Design in Power Electronics

POSITIONS OF RESPONSIBILITY

Reviewer for IEEE Conferences

- ICPS 2025, ECCE Europe 2026, PEDES 2026